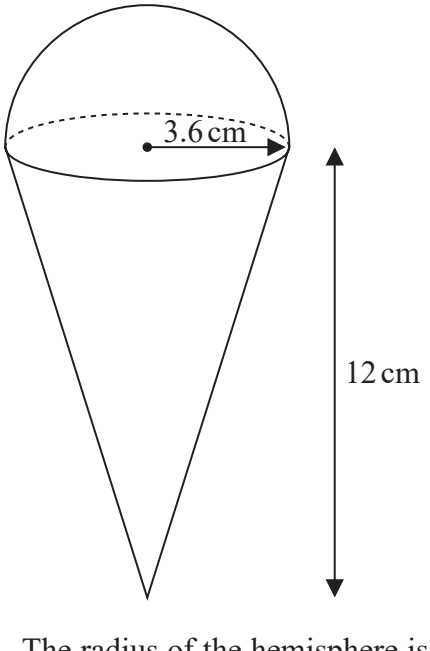
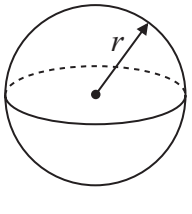
16 Jeff makes two model ice creams, **A** and **B**.

Model **A** is made of a solid hemisphere on top of a solid cone as shown below.



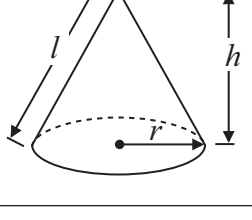
Surface area of sphere = $4\pi r^2$

Volume of sphere = $\frac{4}{3}\pi r^3$



Curved surface area of cone = πrl

Volume of cone = $\frac{1}{3}\pi r^2 h$



The radius of the hemisphere is 3.6 cm.
The radius of the base of the cone is 3.6 cm.
The height of the cone is 12 cm.

Model **A** and model **B** are mathematically similar.

Model **B** is such that

height of model **A** : height of model **B** = 2 : 3

Work out the volume of model **B**.
Give your answer correct to 3 significant figures.

..... cm^3

(Total for Question 16 is 4 marks)